

YUJIN HAM

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RESEARCH INTERESTS

My research interests are in computer vision and deep learning, with a focus on generative video editing for dynamic real-world environments, large-scale 3D/4D reconstruction, and egocentric visual data.

EDUCATION

Rice University Ph.D. in Electrical and Computer Engineering Advisor: <i>Prof. Guha Balakrishnan</i> ↗	Houston, TX, United States Aug 2022 – Expected 2027
Ewha Womans University M.S. in Electronic and Electrical Engineering - Advisor: <i>Prof. Je-Won Kang</i> ↗ - Thesis: Quality-adaptive Image Compression Artifact Removal using Deep Learning	Seoul, South Korea Mar 2020 – Feb 2022
Ewha Womans University B.S. in Electronics and Electrical Engineering	Seoul, South Korea Mar 2016 – Feb 2020

RESEARCH EXPERIENCE

Rice Vision and Imaging Group (RVIG) Advisor: <i>Prof. Guha Balakrishnan</i> ↗ Research Assistant, Rice University - Developed a generative video editing framework for removing people from egocentric walking-tour videos and synthesizing human-free environment walkthroughs; published as first author at CVPR 2026 - Developed DRAGON, a Gaussian Splatting-based registration framework integrating aerial and ground imagery for 3D building reconstruction; published as first author at ICCP 2024 (Oral) - Advanced large-scale, multi-altitude 3D reconstruction for the IARPA WRIVA project through perceptual regularization and diffusion-based refinement	Aug 2022 – Present
Samsung Research America (SRA) Mentor: <i>Abhinav Kumar</i> ; <i>Ph.D.</i> ↗ Research Intern, Mobile Processor Innovation (MPI) Research areas: AI-based computational imaging for video, generative models; work resulted in a patent application	May 2026 – Aug 2026
Information Coding and Processing Lab. (ICPL) Advisor: <i>Prof. Je-Won Kang</i> ↗ Research Assistant, Ewha Womans University Research areas: compression artifact reduction, image enhancement, deep learning, domain adaptation	Mar 2020 – Feb 2022

PUBLICATIONS

- Y. Ham**, J. Kim, V. Boominathan, G. Balakrishnan, “Generating Humanless Environment Walkthroughs from Egocentric Walking Tour Videos”, *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. [↗](#)
- J. Murrugarra-Llerena, P. Chitale, Z. Liu, K. Ao, **Y. Ham**, G. Balakrishnan, P. Cascante-Bonilla, “EgoGroups: A Large-Scale Dataset for Detecting Groups of People in the Wild”, *arXiv preprint arXiv:2603.22249*, 2026.
- Y. Ham**, M. Michalkiewicz, G. Balakrishnan, “DRAGON: Drone and Ground Gaussian Splatting for 3D Building Reconstruction”, *International Conference on Computational Photography (ICCP)*, 2024. [↗](#)
- Y. Ham**, C. Yoo, J. Kang, “Training Compression Artifacts Reduction Network with Domain Adaptation”, *Applications of Digital Image Processing XLIV (SPIE)*, 2021. [↗](#)
- Y. Ham**, J. Kang, “Mid-view Quantization Noise Removal of Multi-view Video using Convolutional Neural Network”, *The Korean Institute of Broadcast and Media Engineers*, 2020.
- N. Kim, **Y. Ham**, J. Kang, “Effective Video Captioning Algorithm Using Feature Attention Model”, *Image Processing and Image Understanding*, 2020.

PROJECTS

Walk-through Rendering from Images of Varying Altitude (WRIVA) <i>Intelligence Advanced Research Projects Activity (IARPA)</i> ↗ - Research areas: large-scale, multi-altitude 3D reconstruction, novel-view synthesis - Enhanced rendering quality using diffusion-based refinement with scene-matched reference across varying altitudes - Improved 3D reconstruction quality through perceptual regularization using DreamSim	Dec 2022 – Present
Compression and Transmission Technologies for Ultra High Quality Immersive Videos Supporting 6DoF <i>Institute of Information & Communications Technology Planning & Evaluation (IITP)</i> ↗ Research areas: DL-based image denoising, 6DoF immersive video, quantization noise removal	Mar 2020 – Dec 2020
A Convolutional Neural Network based Classification System for Educational Learning States using Pupil Sizes <i>Korea Center for Women in Science, Engineering, and Technology (WISET)</i> ↗ Research Areas: ML for Biomedical data classification	Mar 2018 – Dec 2018 <i>Seoul, South Korea</i>

SKILLS

Programming/Tools: Python, C++, MATLAB, LaTeX, Git, Docker
ML/DL: PyTorch, TensorFlow, Hugging Face, Diffusers
Computer Vision/3D: OpenCV, COLMAP, Nerfstudio, 3D Gaussian Splatting
Languages: English (fluent), Korean (native)

HONORS AND AWARDS

Ignite Entrepreneurship Trek to Silicon Valley ↗ <i>Rice University</i>	2025
SWITCH Mentoring Scholarship ↗ <i>Rice University</i>	2024
ICCP 2024 Student Travel Award <i>US National Science Foundation (NSF) & Swiss NSF</i>	2024
Rice University Department of Electrical and Computer Engineering Fellowship <i>Rice University</i>	2022
Grace Hopper Celebration (GHC) Scholarship 2021 <i>AnitaB.org</i> ↗ & <i>Association for Computing Machinery (ACM)</i>	2021
Admission Scholarship for Outstanding Scientists (full tuition for one year) <i>Ewha Womans University</i>	2020
Student Portfolio Award (1st prize) <i>Accreditation Board for Engineering Education of Korea (ABEEK)</i>	2019
Ewha Career Design Scholarship (Merit-based) <i>Ewha Womans University</i>	2019
Dean's List <i>Ewha Womans University</i>	Spring / Fall 2019
Engineering Leadership Scholarship (Merit-based) <i>Ewha Womans University</i>	2018, 2019
Peer Instructor Scholarship (Merit-based) <i>Ewha Womans University</i>	Spring 2018, Fall 2019
Global Frontier Travel Grant (Visiting the US for 2 weeks) <i>Ewha Womans University</i>	2018

TEACHING EXPERIENCE

SWITCH Graduate Mentor <i>Rice University</i> <i>Summer Web-Based Institute for Technologies in CompSci and Healthcare (SWITCH)</i> ↗ Instructor: Prof. Ashutosh Sabharwal	Summer 2024
Teaching Assistant <i>ELEC 542: Generative AI for Image Data (Rice University)</i> Instructor: Prof. Guha Balakrishnan	Fall 2023 / Fall 2026
<i>30266-01: Digital Engineering (Ewha Womans University)</i> Instructor: Prof. Su-Hyun Park	Spring 2021
<i>35477-01: Random Process (Ewha Womans University)</i> Instructor: Prof. Nak-Myeong Kim	Fall 2020
Undergraduate Peer Instructor <i>Ewha Womans University</i> Instructor: Prof. Hye-Sook Lim (<i>Digital Engineering</i>) / Prof. Hyung-Gon Park (<i>Random Process</i>)	Spring 2018 / Fall 2019